

Heat conductance and diffusion equation

diffusion eqns. have the form

$$\frac{\partial u}{\partial t} = D \cdot \Delta u$$

diffusion
constant

$$[D] = \left[\frac{m^2}{s} \right]$$

Laplace
operator

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

they have some conservation law in the background ...

Heat conductance (in solids)

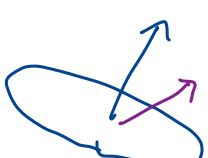
energy density: $\epsilon(x, y, z, t)$

$$[\epsilon] = \left[\frac{J}{m^3} \right] \quad \text{in 3 dimensions}$$

remark: $\epsilon = \epsilon(x, y, z, t)$!

energy current density

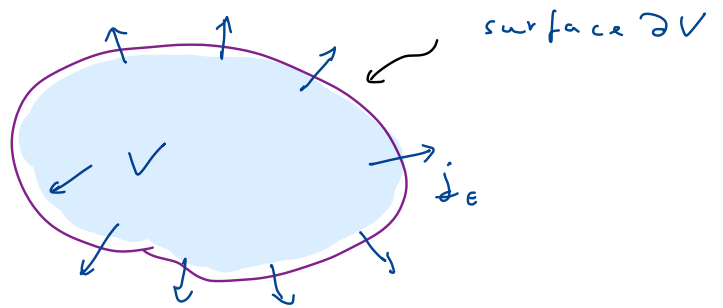
$$j_E = \frac{\text{energy}}{\text{surface} \cdot \text{time}} \quad [j_E] = \left[\frac{J}{m^2 \cdot s} \right] \quad \text{in 3 dim.}$$



dA surface element
energy through this / s = $d\vec{A} \cdot \vec{j}_E$

remark $j_E(x, y, z, t) \sim$ vector field
($\sim$ electric field)

consider
volume V
its surface ∂V



energy in V

$$E = E(t) = \int_V dx dy dz E(x, y, z, t)$$

$$\text{energy out / sec} = \int_{\partial V} d\vec{A} \cdot \vec{j}_E(x, y, z, t)$$

$$\frac{dE}{dt} = - \frac{\text{energy out}}{\text{sec}} = - \int_{\partial V} d\vec{A} \cdot \vec{j}_E(x, y, z, t)$$

but

$$\frac{dE}{dt} = \int_V dV \frac{\partial E(x, y, z, t)}{\partial t}$$

and

$$\int_{\partial V} d\vec{A} \cdot \vec{j}_E(x, y, z, t) = \int_V dV \operatorname{div} \vec{j}_E(x, y, z, t) \quad \text{Gauss !!}$$

$$\text{here } \vec{j}_E = \begin{bmatrix} j_E^x \\ j_E^y \\ j_E^z \end{bmatrix} = \frac{\partial E}{\partial x} \hat{x} + \frac{\partial E}{\partial y} \hat{y} + \frac{\partial E}{\partial z} \hat{z}$$

Therefore

$$\int_V dV \left(\frac{\partial E}{\partial t} + \operatorname{div} \vec{j}_E \right) = 0$$

since this holds for $\forall$ volume $V \Rightarrow$
integrand must vanish!

(2)

$$\frac{\partial \epsilon}{\partial t} + \operatorname{div} \mathbf{j}_\epsilon = 0$$

Continuity
equation
~ energy conser-
vation

both ϵ and $\mathbf{j}_\epsilon$ are related to temperature, u :

- $\epsilon \sim u = T - T_0$
- $\mathbf{j}_\epsilon \sim -\operatorname{grad} u$

- energy density

$$(a) \quad \epsilon = c \cdot \rho \cdot u$$

specific
heat

$$[c] = \left[\frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}} \right]$$

density

$$[\rho] = \left[\frac{\text{kg}}{\text{m}^3} \right]$$

temperature w.r. t.
some reference T_0

$$[u] = [^\circ\text{C}]$$

- energy current from hot to cool place:

$$(b) \quad \mathbf{j}_\epsilon = -\kappa \cdot \operatorname{grad} u$$

heat conductance

$$[\kappa] = \left[\frac{\text{J}}{\text{m} \cdot \text{s} \cdot ^\circ\text{C}} \right] = \left[\frac{\text{kg} \cdot \text{m}}{\text{s}^3 \cdot ^\circ\text{C}} \right]$$

substitute a, b to (2):

$$c \cdot \rho \frac{\partial u}{\partial t} - \kappa \operatorname{div} \operatorname{grad} u = 0$$

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

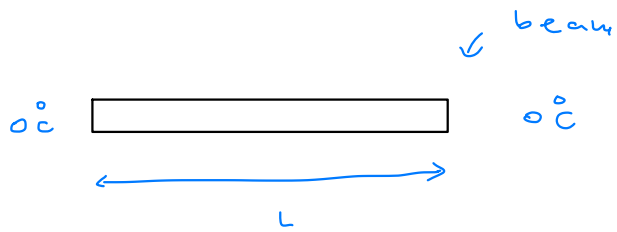
$$\frac{\partial u}{\partial t} = D \cdot \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \quad D = \frac{\kappa}{c \cdot \rho}$$

Solution in one dimension

$$\frac{\partial u}{\partial t} = D \cdot \frac{\partial^2 u}{\partial x^2}$$

boundary condition:

Dirichlet - cond



$$u(0, t) = u(L, t) = 0$$

$$u(x, t=0) = f(x) \leftarrow \text{initial profile temperature}$$

Neumann condition

and is insulating! $\Rightarrow f_e(0) = f_e(L) = 0$

$$\Rightarrow \partial_x u(0, t) = \partial_x u(L, t) = 0$$

separation of variables:

$$u(x, t) = F(x) \cdot G(t) \quad \Rightarrow \quad \frac{F''(x)}{F(x)} = \frac{\dot{G}(t)}{G(t)} \cdot \frac{1}{D} = \lambda$$

Dirichlet:

$$F(0) = F(L) = 0 \quad \Rightarrow \quad 0 > \lambda = -p^2 \quad \Rightarrow$$

$$\Rightarrow F_n(x) = \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) \quad \Rightarrow \quad \lambda = -\frac{n^2 \cdot \pi^2}{L^2}$$

$$\Rightarrow \frac{\dot{G}}{G} = -D \cdot \frac{n^2 \pi^2}{L^2} \quad \Rightarrow \quad G_n(t) = e^{-D \frac{n^2 \pi^2}{L^2} \cdot t}$$

particular solution:

$$G_n(t) F_n(x) = e^{-D \frac{n^2 \pi^2}{L^2} \cdot t} \cdot \sin\left(\frac{n \cdot \pi}{L} \cdot x\right)$$

$\nearrow$ exponential decay $\nearrow$ mode

characteristic time:

$$t_0 \equiv \frac{L^2}{D \cdot \pi^2}$$

general solution:

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) e^{-n^2 \frac{D \cdot \pi^2}{L^2} \cdot t}$$

A_n 's follow from B.C. at $t=0$

$$f(x) = u(x, 0) = \sum_{n=1}^{\infty} A_n \cdot \sin\left(n \cdot \frac{\pi}{L} \cdot x\right)$$

$$\Rightarrow A_n = \frac{2}{L} \int_0^L dx f(x) \cdot \sin\left(n \cdot \frac{\pi}{L} \cdot x\right)$$

Example:

$t=0$



$$f(x) = \begin{cases} \frac{2u_0}{L} \cdot x & x < \frac{L}{2} \\ \frac{2u_0}{L} \cdot (L-x) & x > \frac{L}{2} \end{cases}$$

$$A_n = \frac{2}{L} \int_0^L dx \sin\left(n \cdot \frac{\pi}{L} \cdot x\right) f(x) = \dots$$

$n \text{ odd} = 2\ell + 1$

$$= \frac{(-1)^\ell \cdot 8}{\pi^2 \cdot (2\ell + 1)^2} \cdot u_0$$

$\ell = 0, 1, \dots$

$$u(x, t) = u_0 \cdot \frac{8}{\pi^2} \left\{ \sin\left(\frac{\pi}{L} \cdot x\right) \cdot e^{-t/t_0} - \frac{1}{9} \sin\left(\frac{3\pi}{L} \cdot x\right) e^{-9 \cdot t/t_0} \right. \\ \left. + \dots \cdot e^{-25 \cdot t/t_0} + \dots \right\}$$

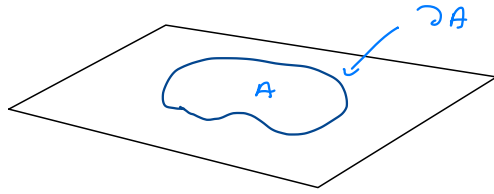
• modes decay exponentially after time $t \approx t_0 = \frac{L^2}{D \cdot \pi^2}$

• high modes decay extremely fast

$$\Rightarrow t \gtrsim t_0 \Rightarrow u(x, t) \approx \frac{8u_0}{\pi^2} \sin\left(\frac{\pi}{L} \cdot x\right) \cdot e^{-t/t_0}$$

Remark: solution is different in case of Neumann condition !! $\Rightarrow$ Tutorial

2-dimensional problem

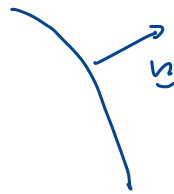


$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$$

boundary conditions:

(a) $u(x, y \in \partial A) = f(x, y) \sim \text{Dirichlet (perfect conductor)}$

(b) $\frac{\partial u}{\partial n} = 0 \text{ at } \partial A \text{ Neumann (insulating)}$

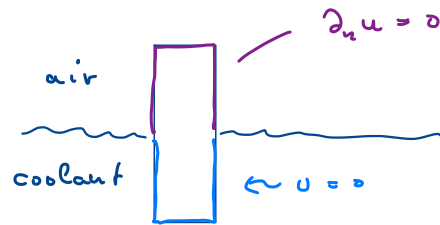


$$\vec{j}_E \sim \vec{\nabla} \cdot u \Rightarrow$$

$$\vec{n} \cdot \vec{j}_E \sim \vec{n} \cdot \vec{\nabla} u = \frac{\partial u}{\partial n} = 0$$

heat cannot flow out...

(c) mixed e.g.



$\Rightarrow$ separation of variables?

$$u(x, y, t) = F(x, y) G(t)$$

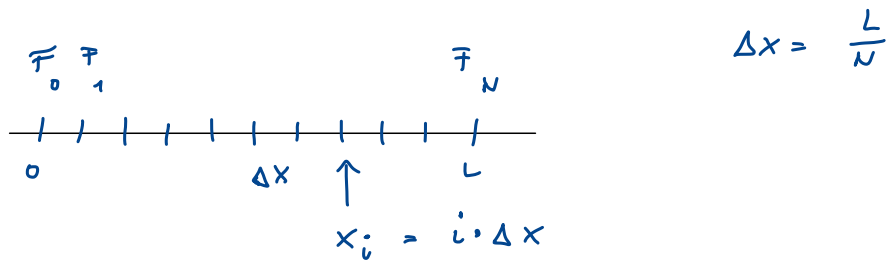
$$\Rightarrow \frac{1}{D} \frac{\dot{G}}{G} = \frac{1}{F} \Delta F \equiv \lambda = -p^2$$

(3) $\Delta F = \lambda F$

find eigenvalues λ and eigenmodes F satisfying (3)

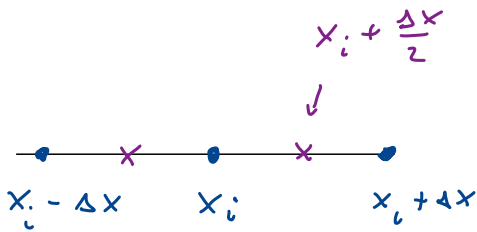
$\Rightarrow$ perform numerically $\Rightarrow$ Finite Element Method

Illustration in 1D



need to determine

$$\left. \frac{d^2 F}{dx^2} \right|_{x_i} \approx ?$$



$$\frac{F_{i+1} - F_i}{\Delta x} \approx F' \left(x_i + \frac{\Delta x}{2} \right)$$

$$\frac{F_i - F_{i-1}}{\Delta x} \approx F' \left(x_i - \frac{\Delta x}{2} \right)$$

$$F''(x_i) \approx \frac{F'(x_i + \Delta x/2) - F'(x_i - \Delta x/2)}{\Delta x} =$$

$$= \frac{1}{\Delta x^2} (F_{i+1} + F_{i-1} - 2F_i)$$

$$F''(x) = \mathcal{H} F(x) \Rightarrow F_{i+1} + F_{i-1} = (2 + \Delta x^2 \mathcal{H}) F_i$$

boundary condition

$$F_N = F_0 = 0 \quad (\text{Dirichlet})$$

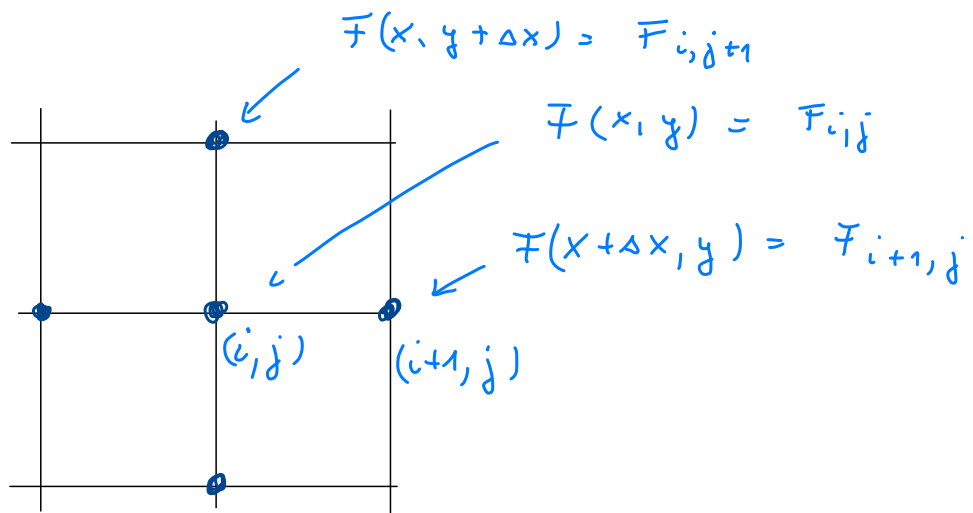
$$\left(F'_0 = 2F_1 - \frac{3}{2}F_0 - \frac{1}{2}F_2 = 0 \quad (\text{Neumann}) \right)$$

$$\begin{bmatrix} -2 & 1 & & \\ & 1 & -2 & 1 \\ & & 1 & \\ & & & \ddots \\ & & & & 1 & -2 \\ & & & & & 1 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ \vdots \\ F_{N-1} \end{bmatrix} = \Delta x^2 \mathcal{H} \begin{bmatrix} F_1 \\ F_2 \\ \vdots \\ F_{N-1} \end{bmatrix}$$

linear equation! $\Rightarrow$ eigenvalues $\sim \Delta x^2 \mathcal{H}$

2D case

(a) rectangular lattice $x = i \cdot \Delta x$, $y = j \cdot \Delta x$



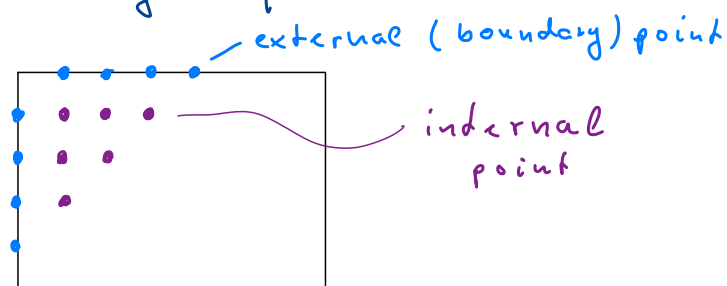
$$\left(\frac{\partial^2 F}{\partial x^2} \right)_{i, j} = \frac{1}{\Delta x^2} (F_{i+1, j} + F_{i-1, j} - 2 F_{i, j})$$

$$\left(\frac{\partial^2 F}{\partial y^2} \right)_{i, j} = \frac{1}{\Delta x^2} (F_{i, j+1} + F_{i, j-1} - 2 F_{i, j})$$

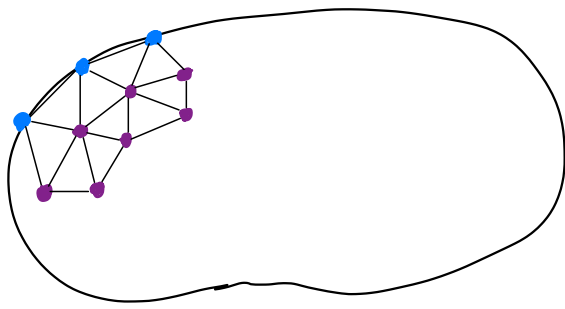
$$\Rightarrow (\Delta F)_{i, j} = \frac{1}{\Delta x^2} \left\{ F_{i+1, j} + F_{i-1, j} + F_{i, j+1} + F_{i, j-1} - 4 F_{i, j} \right\}$$

$$(\Delta F)_{i, j} = 0$$

- linear equ. for internal points
- as many eqs as internal points



(b) general case



triangulation /

- express ΔF at internal point k in terms of its neighbors
- solve linear eqn.
- interpolate solution

Steady state solution (Tutorial)

$$\frac{\partial u}{\partial t} = 0 \Rightarrow$$

$$\Delta u = 0$$

Laplace eqn.

solution for rectangular shape:

⋮